

September Week 1 Electraders

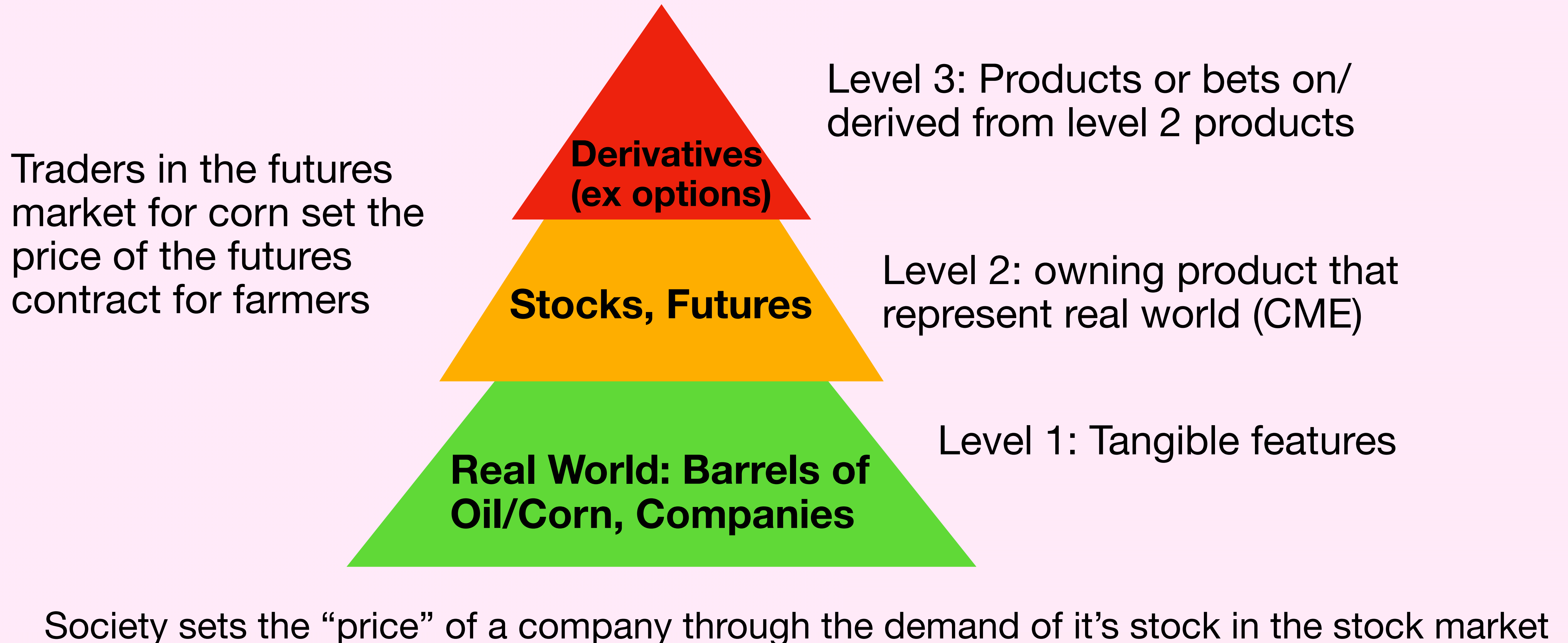
Introduction

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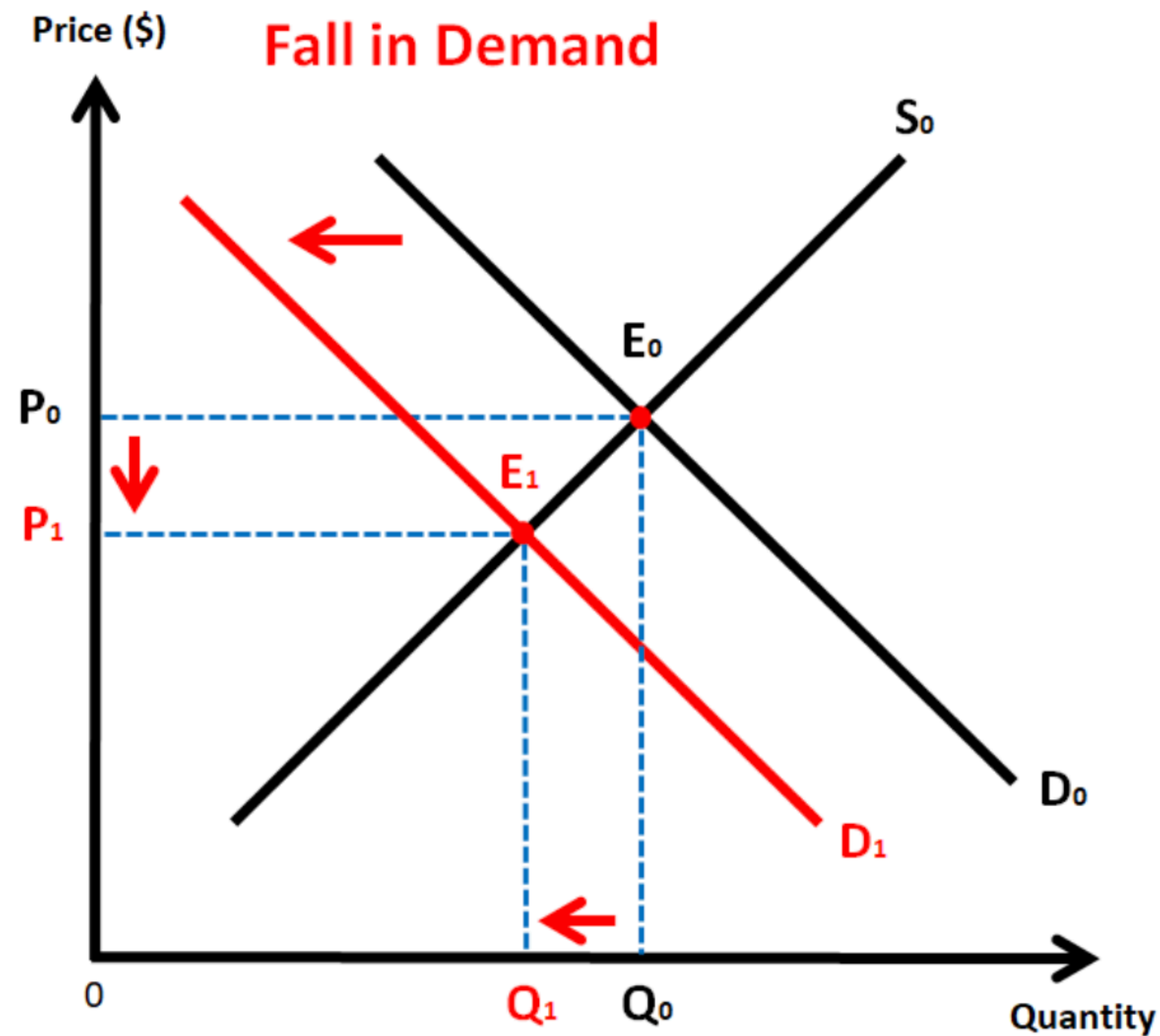
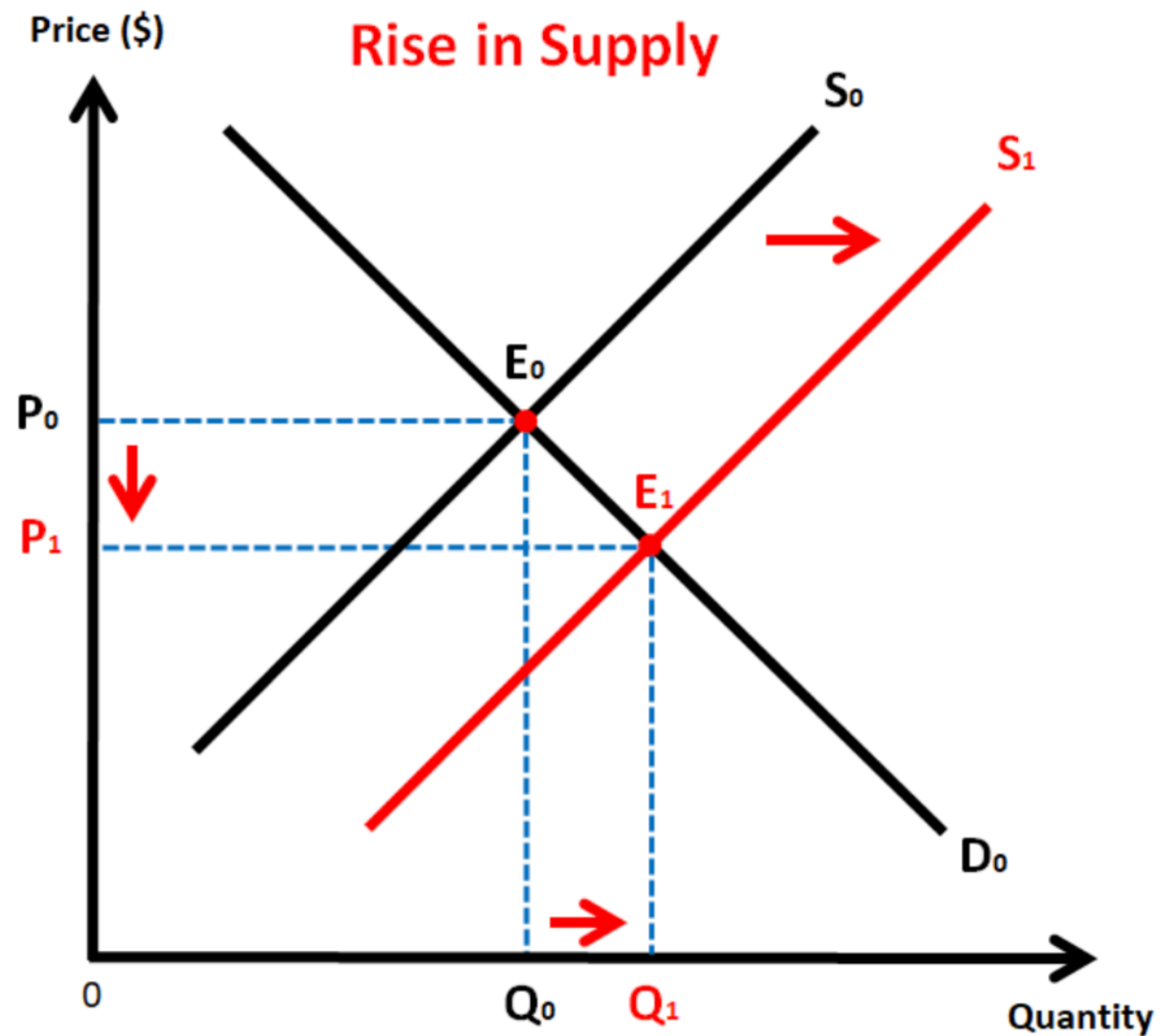
- What are Futures and CME
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Fundamentals (Intro to futures and CME)

Levels of abstractions in finance market



Supply and Demand



What are futures?

- Future contract specify a date, location, and quantity of a good to be delivered. The price which is it sold at changes. At expiry, it equals the local price at location (no arb).
- They are used for **hedging** (unlike stocks) by farmer/companies, and for **speculation** (with extra leverage over stocks) by traders
- **Hedging:** If a farmer wants to guarantee a certain price for their corn sold in a future month, they secure a price with a future contract in case the price drops. Same with Cereal companies buying corn to protect against price rise.
- **Speculation:** If a future contract for corn barrel sells at \$70, buying 1,000 barrels would be expensive. CME allows paying only an **initial margin** ([website link](#)) of about 5% of the purchase for collateral. In this case, by paying \$3,500, we can buy 1,000 barrels and if the price moves from \$70 to \$71 we make \$1,000.
- When trading our gains and loses are returned or taken from us at the end of the day and this is called **Marked to Market**. Our margin can never drop below a set amount.

How CME works

- CME creates legal future contracts and has the responsibility to connect the buyer of contracts with distributors.
- Unlike the stock market, the CME can create unlimited future contracts.
- However, only **Hedgers** will hold on to their contract until expiry since they actually have the good to sell or have use in owning the good.
- If I decide to sell a future (without ever owning one), CME will match me with someone willing to buy. If the contract decreases by \$100 in price, the buyer will just pay me \$100 (similar to just a horse race bet).
- I can exit my short (sell) position by buying the exact same contract, CME will cancel (clear) all my obligations. If I don't exit my contract by expiry, I will (hypothetically) be forced to somehow deliver a barrel of corn to the location specified in the contract.
- This is why about 95% of future contracts are closed before delivery.

Team Logistics

Type of trading

- For purposes of CME we can do **technical** or **fundamental** trading- in reality there is also **quant** trading which uses algorithms and high performance computers to capture statical fluctuations in the market. The algorithms also includes fundamental and technical analysis.
- **Fundamental analysis** looks into real world features of a company or farm/oil yield to determine if the price of the stock or the future is accurate, if not what is the movement. For example, a country announces they will produce more oil next week or it rained alot this season, this affects the supply oil/corn in the following months and therefore the future price.
- **Technical analysis** looks for trends in the stock price to detect if demand is changing and thus if the price will rise or fall. There are equations which lead to indicators if this the case. Many famous indicators are built into our CQG trading platform, thus the matter comes down to choosing 3 or so indicators which give the best predictions for a given product. We can look for assets which have similar price trends and thus will share the same indicators.

Asset classes and team trading

- Agriculture
- Energy
- Metals
- Equity Index
- Interest Rates
- Foreign Exchange

- Different asset classes have their own features
- Ex. Equities are the less reactive to news and most reliable for technical trading
- Suggested team: 2 strategy creators, 2 trade executors, 1 risk manager.
- Simple risk management: Stop at 1% increase and .5% decrease
- Ex: If we have 500,000, leave strategy when gain 5,000 or lose 2,500.
- An Alternative: Everyone trades one day a week, or 4 traders + 1 reporter
- Suggested plan this week: Everyone comes up with their own strategy and tests it
- Alternative plan: Assign specific technical indicators for each person to specialize in and we can dissect the parts of the strategy (more later)

Momentum Strategy (crude oil)

Momentum: $M_t^n = P_t - P_{t-n}$, where P_t is the time t price and n is n -bars ago.

Strategy: If $M > 0$ go long (buy)
If $M < 0$ go short (sell)

Red down arrows indicate $M < 0$, blue up indicate $M > 0$

Works because it captures trends!

$n=20$:



$n=5$:



For $N=20$, this behavior is known as a chop: (revisit later)



Simple Moving Averages

- Momentum reacts quickly, we can smooth out the reaction and see chops with **moving averages**. First, lets consider **simple moving average**.

$$SMA_t(n) = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i}$$

Strategy: For $n > n'$, buy when $SMA(n) > SMA(n')$, else sell



Ex, In the graph:

$SMA_t(20) > SMA_t(50) \rightarrow long$

$SMA_t(50) > SMA_t(20) \rightarrow short$

Why $n=20$ and 50 ? Lowers noise as,
 $20 \times 5 = 100$ min captures short trend
 $50 \times 5 = 250$ min captures mid-day trend

This works because if a short term trend is greater than a longer one, it implies momentum builds, and vice versa.

**Chop when
Slope ≈ 0 :**



Moving Averages + Momentum to avoid chop

- For moving averages we have seen that the slope near zero indicates choppy behavior, during which the MA strategies are not useful.
- A slope relationship between the two $\frac{1}{n}M_t^{20} = SMA_t(20) - SMA_{t-1}(20)$
- This is still a bit hard to predict, combining momentum and MA gives cleaner signals

Idea:

Buy when $M_t^{20} > 0$
and $P_t > SMA_t(50)$

Sell when $M_t^{20} < 0$
and $P_t < SMA_t(50)$

Also, no trades
when Slope ≈ 0



Chop Resolution (EMA + momentum)

- Similar but the exponential weighed average gives more importance to recent prices. Same rule as before with SMA<-> EMA.



EMA is defined by:

$$EMA_t = \alpha P_t + (1 - \alpha) EMA_{t-1}$$

Where $\alpha = \frac{2}{N + 1}$ and the

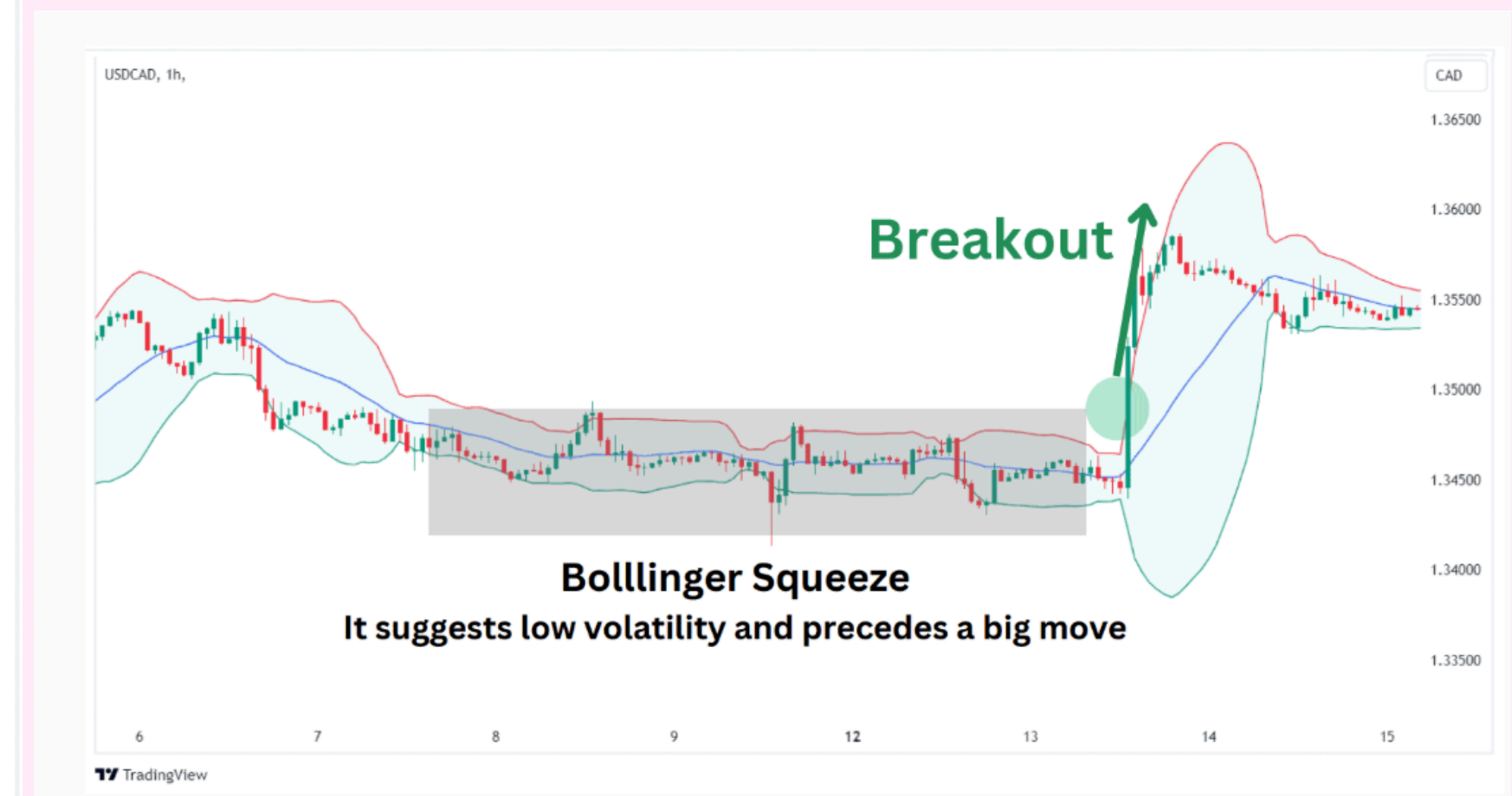
initialization :

$$EMA_0(N) = SMA_0(N)$$

Intro to Bollinger Bands: Predicting Reversion

- There are strategies to predict if an upward trend goes down again (reverts)
- There are 3 Bollinger Bands, middle: $SMA(20)$ and upper/lower: $SMA(20) \pm 2\sigma$

See for strategies: <https://alchemymarkets.com/education/indicators/bollinger-bands/>



Strategies to try

- For momentum, combining $n=5, 10, 15/20$ and only trading when they are all $M>0$
- Combinations of Moving Averages/ crossings
- Moving Averages + RSI/ Momentum
- Bollinger Bands+ MACD/ Keltner Channel/ ATR: Squeezes/ Breakouts
- Bollinger Bands+ RSI/ SMA (50,200): reversals or breakout
- MACD strategies (combine with others)
- $MACD_t = EMA_t(12) - EMA_t(26)$
- See JJ2024.ipyb and JJ2024 paper to see more complicated technical indicators and how to backtest using python

Macro (or fundamental) analysis example (old)

- We have been looking at how to predict crude oil future-price movements using technical analysis.
- However, new news or information causes traders to spectate (buy/sell) in a certain way depending if this causes them to believe the price will rise or fall.
- For example, OPEC+, an organization of major oil exporting companies announced more oil production, we see from this website that

After falling about 3% last week, oil prices **↑ CL.1 +0.47%** rose 1% after OPEC+ agreed Sunday to increase oil production in October, maintaining its months-long output boost in an effort to regain market share at the expense of lower prices.

- By following the news we can act quickly to catch trends before our technical indicators. We should take this approach into consideration.

Official Week 1 approach (old)

- Since we don't have much experience I believe a reasonable approach will be to pick one commodity to focus on (say crude oil or corn), and use it to understand different strategies. Its also important to start develop a working rhythm with each other to best prepare for the competition.
- I suggest we start with either 1 macro analyst + 3 technical traders+ strategy coordinator.
- The traders can experiment with different windows, one can focus on only MA, another only momentum, and a third on MA + momentum. Alternatively we can have MA/ momentum, MA+momentum, and MA+momentum+ volatility
- The strategy coordinator can review the work by each trader and determine/report an analysis of each traders strategy- best/worst/ what worked and didn't.
- If we don't want to do fundamental analysis yet, we can do 4 traders this week to get rolling and adapt our strategy for week 2.